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Faculty of Exact Sciences and Computer Science
Department of Computer Science



Capstone Project (PFC)

Design and Implementation of a Document Management System

Prepared by:

Ms. FETHALLAH Hiba
Ms. NIATI Ayaterrahmane
Ms. MENASRI Ibtihel
Ms. BENNANI Sarah

Supervised by:

Mr. BELHOCINE Farouk
Mrs. DELLAL-HEDJAZI Badiâa

Board of Examiners:

Mrs. ABDAT Nadia *President* (USTHB)

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ABSTRACT

As digital transformation accelerates, organizations require robust, scalable solutions to manage the exponential growth of electronic documents. For **ITComp**, serving over 700 international clients, the reliance on third-party integrations presented significant licensing costs and technical limitations. To address this, this graduation project details the design and implementation of a proprietary, enterprise-grade **Electronic Document Management System (EDMS)**.

The primary objective was to engineer a **cloud-native, Software as a Service (SaaS)** platform built upon a strict **multi-tenant architecture**. Developed using **Domain-Driven Design (DDD)** principles with **ASP.NET Core, the ABP Framework, and React**, the system ensures complete logical data isolation between distinct client organizations. To guarantee absolute data privacy and governance, a custom **"3D Security Matrix"** was implemented. This advanced authorization engine seamlessly combines global **Role-Based Access Control (RBAC)**, **item-level Access Control Lists (ACLs)**, user clearance levels, and a document lifecycle **state machine**.

Crucially, as a **100% locally developed solution**, the platform guarantees **national data sovereignty**. It ensures that sensitive corporate information remains strictly within **Algerian borders**, providing full compliance with local data protection laws (**Law 18-07**) and bypassing the risks of foreign cloud hosting. Furthermore, the integration of automated **OCR** and **Elasticsearch** indexing provides high-performance data retrieval. The resulting **modular software block** not only eliminates third-party dependencies for ITComp but also delivers a highly secure, scalable, and customizable document engine ready to be deployed across its global B2B ecosystem.

Keywords: Electronic Document Management System (EDMS), SaaS, Multi-Tenancy, ABP Framework, React, Access Control List (ACL), Domain-Driven Design (DDD).

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Table of Acronyms

Acronym	Signification
ABP	ASP.NET Boilerplate / Advanced Business Platform
ACL	Access Control List
AI	Artificial Intelligence
API	Application Programming Interface
B2B	Business-to-Business
B2C	Business-to-Consumer
DAM	Digital Asset Management
DDD	Domain-Driven Design
DMS	Document Management System
ECM	Enterprise Content Management
EDMS	Electronic Document Management System
ERP	Enterprise Resource Planning
HTML	HyperText Markup Language
IDP	Intelligent Document Processing
ISO	International Organization for Standardization
IT	Information Technology
JWT	JSON Web Token
LTS	Long Term Support
ML	Machine Learning
OCR	Optical Character Recognition
OS	Operating System
PAM	Privileged Access Management
PDF	Portable Document Format
RBAC	Role-Based Access Control
REST	Representational State Transfer
SaaS	Software as a Service
SDK	Software Development Kit
SQL	Structured Query Language
UI/UX	User Interface / User Experience
UML	Unified Modeling Language
VM	Virtual Machine

General Introduction

Context of the Study

In the era of the Fourth Industrial Revolution, the digitisation of information has become a vital strategic pillar for any organization seeking efficiency and competitiveness. The exponential growth of digital data volume forces companies to rethink their storage, classification, and retrieval methods. This digital shift, while offering immense opportunities for collaboration, also brings critical challenges regarding data governance, security, and long-term preservation.

Within the Algerian economic landscape, this transformation is further accelerated by a growing need for **Digital Sovereignty**. Organizations are increasingly seeking local alternatives to international solutions to ensure that their sensitive corporate data remains strictly within national borders, in full compliance with local regulations such as **Law 18-07** on personal data protection.

Problem Statement

ITComp, a leading player in digital transformation with over 700 international clients, currently integrates third-party solutions to provide document management features. However, this dependency leads to significant recurring licensing costs and technical constraints that limit functional customization.

The core challenge of this project is to eliminate this dependency by developing a proprietary **Electronic Document Management System (EDMS)**. This system must not only match the performance of market leaders but also operate as a modern **SaaS (Software as a Service)** platform. The primary difficulty lies in designing a strict **multi-tenant architecture** capable of ensuring absolute data isolation, alongside a granular security model that adapts to the complex organizational structures of ITComp's diverse client base.

Objectives of the Project

The main goal of this graduation project (PFE) is to design and implement a robust, scalable, and secure EDMS engine. The specific objectives are:

- **Technical Independence:** Developing a modular software block that serves as an internal document engine for ITComp's broader product suite.

- **High-Level Security:** Implementing a custom "3D Security Matrix" combining Role-Based Access Control (RBAC), item-level Access Control Lists (ACLs), and user clearance levels to ensure military-grade data protection.
- **Advanced Processing:** Integrating Optical Character Recognition (**OCR**) and **Elasticsearch** for automated metadata extraction and near-instantaneous full-text search.
- **SaaS Optimization:** Leveraging the **ABP Framework** and **React** to manage a high-performance multi-tenant environment with optimized infrastructural deployment.

Report Organization

This capstone project report is organized into three main chapters, detailing the progression from theoretical analysis to technical realization:

- **Chapter 1: State of the Art and Requirement Analysis** – This chapter explores the theoretical foundations of EDMS and SaaS architectures. It details the functional and non-functional requirements and defines the project scope through UML use case modeling.
- **Chapter 2: System Design and Architecture** – This section focuses on the technical design of the solution. It covers the Domain-Driven Design (DDD) approach, database modeling, the multi-tenant security strategy, and the decoupled system architecture.
- **Chapter 3: Implementation and Realization** – The final chapter details the development environment and the technologies used. It showcases the implementation of key features, such as the OCR pipeline, and presents the final interfaces of the application.

The report concludes with a general assessment of the work achieved and discusses potential future improvements and perspectives for the system.

Chapter 1

State of the Art and Requirement Analysis

1.1 Introduction

This chapter establishes the complete context of the project. It begins with a presentation of the host organization, ITComp, and the core problem statement. Following this, it provides a theoretical overview of Document Management Systems (DMS) and their essential components, introduces the key underlying technologies, and concludes with a comparative state-of-the-art review of existing solutions.

1.2 Hosting Company: ITComp

1.2.1 Presentation of the host organization

ITComp is a leading digital transformation firm founded in **1999**. With over 25 years of experience, the company specializes in IT services, professional training, and digital solution integration. While headquartered in **Algeria**, ITComp maintains a strong international presence, having delivered projects across **18 French-speaking African countries**, France, Luxembourg, and Belgium.

The organization serves a diverse portfolio of over **700 clients**, positioning itself as a key player in the Euro-African technology landscape.



Figure 1.1: ItComp Company Logo.

1.2.2 Missions of the host organization

The core mission of ITComp is to support businesses in their technological evolution through four strategic pillars:

- **Integration and Development:** Modernizing information systems, data migration, and custom application development.
- **Professional Training:** Providing over 300 certification programs (Cloud, AI, Cybersecurity, DevOps) as a B2B and B2C leader.

- **Technical Assistance:** Ensuring infrastructure stability through reactive support and preventive maintenance.
- **Strategic Consulting:** Guiding digital transformation through tailored technological roadmaps.

1.2.3 Structure and services offered

ITComp is organized into four specialized divisions, each focusing on a distinct area of the IT lifecycle:

1. **ITComp Engineering:** Focuses on complex infrastructure, servers, and virtualization.
2. **ITComp Solutions:** Dedicated to software development, ERP, Big Data, and Artificial Intelligence.
3. **ITComp Education:** An authorized testing and training center (the only Oracle University certified center in Algeria).
4. **ITComp Innovation:** A specialized hub for creating and developing home-grown digital solutions in Algeria.

1.3 Strategic Needs and System Requirements

To support its international growth and reduce third-party licensing costs, ITComp aims to develop a proprietary, high-performance Electronic Document Management System (EDMS). This strategic shift towards technical independence dictates the following core system requirements:

1.3.1 Functional Requirements

- **Access and Collaboration:** Manage complex organizational structures using traditional Role-Based Access Control (RBAC) combined with dynamic Attribute-Based Access Control (ABAC) via tags. This ensures scalable, secure, and trackable document sharing across large departments.
- **Automated Capture & Search:** Integrate an Optical Character Recognition (OCR) engine for text digitization, coupled with Elasticsearch for near-instant, full-text retrieval across massive data volumes.
- **Document Organization:** Enable intuitive document management through hierarchical folders and dynamic, customizable metadata tagging.

1.3.2 Non-Functional Requirements

- **SaaS Multi-Tenancy:** Operate a highly cohesive, cloud-native instance capable of serving multiple clients simultaneously while guaranteeing absolute logical data isolation.
- **Digital Sovereignty:** Utilize a decoupled storage architecture to support localized physical hosting, ensuring strict compliance with Algerian data protection regulations (Law 18-07).
- **Ecosystem Modularity:** Function as a containerized, "plug-and-play" software block that seamlessly integrates into ITComp's broader product catalog.
- **Traceability & Usability:** Deliver a responsive, role-specific user interface backed by an immutable audit trail logging all critical system actions.

1.4 Document Management System

A document management system (DMS) is software that creates, stores, manages, and tracks electronic documents and images of paper-based documents. A DMS can be employed to capture, distribute, and track documents. It can also be used to manage channel workflows, output systems, and information retrieval systems, making the capturing, digitizing, and tagging of documents quick and easy. [web1]

Document management is defined as the control and coordination of all the processes involved in the flow of paper and electronic documents to ensure that authorized personnel have secure and efficient access to the documents whenever the need arises. The processes involved in the flow of a document include document capture, storage, indexing, routing, distribution, and retrieval.[1]

1.4.1 Use Cases for Document Management Systems

Document management systems are used to capture and organize all types of documents into an electronic form that is easy to find and retrieve when needed. A document in this case, as defined by the ISO 12651-2[1], is any recorded object or information that is considered a unit of data. A paper document, such as an invoice, is first scanned and integrated into the document management system in electronic format for future retrieval [1].

The document is then indexed by categories such as department, order number, vendor, or any other identifiable concept relevant to your business. Unlike traditional

[1]ISO (the International Organization for Standardization) is a worldwide federation of national standards bodies. ISO collaborates closely with the International Electrotechnical Commission (IEC) on all matters of electrotechnical standardization.

file cabinet folders, a document management system stores documents in an electronic format that is easier to be accessed by authorized personnel and customers [1].

1.5 Key Components

1.5.1 Document Metadata

Each document is usually stored along with its metadata, such as the date of storage and the identity of the person storing the document. The system can take metadata automatically or request the user to provide it.[web1]

1.5.2 Document Capture

Capture in document management refers to the processing of paper documents through scanning to enable storage in electronic format. [web1]

1.5.3 Storage

Storage involves how the document is kept in a central repository for later retrieval. This includes specific locations where each document is stored, retention periods, migration through different storage media, and the eventual destruction of a document. [web1]

1.5.4 Document Retrieval

Retrieval is a key element of every document management system. It is a complex process that regulates permissions and access, allowing users to find files via unique identifiers like metadata or specific search terms.[web1]

1.5.5 Searching

Searching allows users to find specific documents and folders using template attributes or via a full-text search. [web1]

1.5.6 Sharing and Distribution

Regulated sharing allows the distribution of documents, sections, or entire accounts with authorized individuals. [web1]

1.5.7 Document Security

With advanced security threats such as crypto-ransomware [2], it is now mandatory to ensure that your documents are stored in a safe and secure central repository. [web1]

1.6 Docker

Docker is a container virtualization technology, acting as a lightweight virtual machine (VM). It provides a developer workflow that helps teams build applications inside isolated containers and share them efficiently [2].

Docker Images and Container

Containers hold the entire kit required for an application to run in isolation. Images act as read-only templates to create these containers. Changes are made by "committing" to a base image or by using an automated "Docker build" command with a Dockerfile.[3]

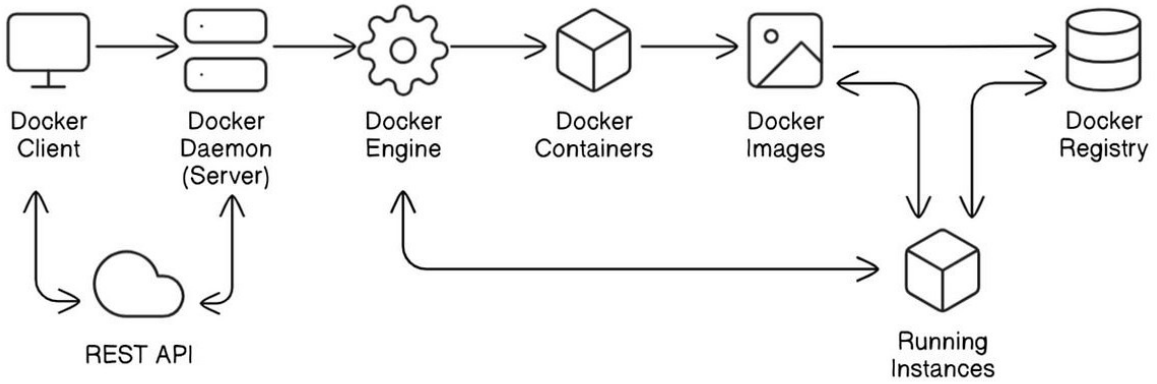


Figure 1.2: Standard Client-Server Architecture of the Docker Engine [web2]

1.7 Optical Character Recognition (OCR)

Optical Character Recognition (OCR) enables scanned documents to become fully searchable documents with text content recognized by computers. It extracts relevant information and automatically enters it into electronic databases, replacing manual retyping [4].

[2]Crypto-ransomware is a type of malware that encrypts a victim's files, making them inaccessible until a ransom is paid.

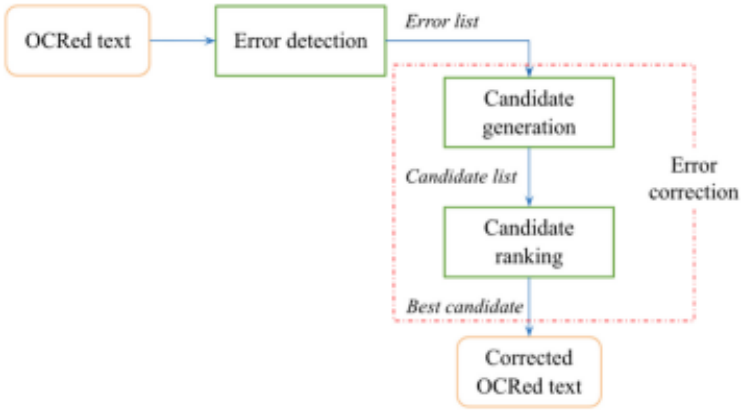


Figure 1.3: Post-OCR processing pipeline.[5]

1.8 Elastic Search

Elasticsearch is an open-source, distributed search and analytics software built on Apache Lucene. It's designed for horizontal scalability, reliability, and real-time search. Elasticsearch enables complex search capabilities across diverse data types and is widely recognized for its powerful full-text search, structured search, analytics capabilities, and ease of use.[6]

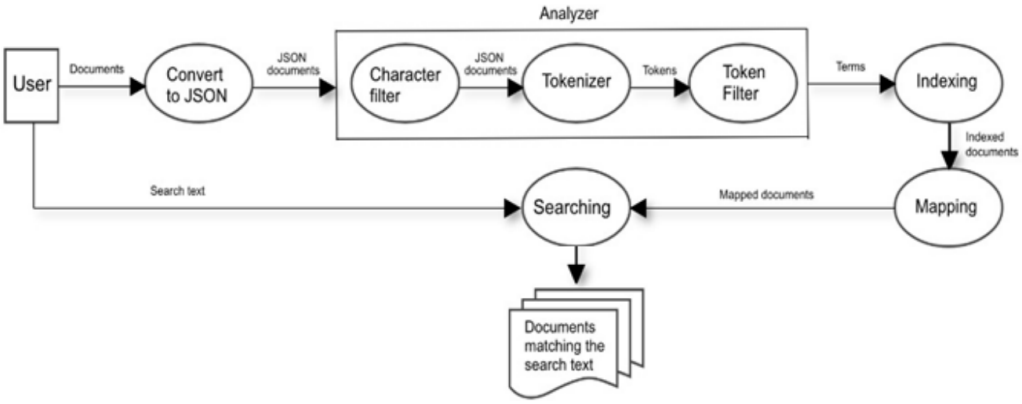


Figure 1.4: Text searching process in Elasticsearch [7]

1.9 Review of Existing Solutions

1.9.1 Proprietary & Ecosystem Leaders

Microsoft SharePoint

SharePoint [3] is a comprehensive, purpose-built Document Management System that offers all core DMS functionality with exceptional integration across Microsoft 365 . For organisations already using Teams and other Microsoft technologies, it eliminates technology silos and significantly improves employee experience. SharePoint Premium adds intelligent document processing and AI-powered content classification, unlocking value from the 80% of enterprise data that remains unstructured.[web3]

M-Files

M-Files [4] is a metadata-driven document management system that helps business and IT leaders work faster and smarter [web4]. Documents are easy to find and trust, and the system is fully ready to support AI integration.

DocuWare

DocuWare IDP [5] leverages best-in-class AI to provide customers with the fastest document lifecycle automation rates, greatly improving overall productivity and earnings [web5]. Its key features include:

- Automatic classification that triggers document workflows.
- Automation for almost any document type, including weathered scans or handwriting.
- Precise extraction of complex data positions regardless of page breaks.
- A trainable AI model for custom workflows and accurate data from the start.
- Pre-trained use cases for reduced configuration requirements.
- Background processing that facilitates less manual intervention.
- Ability to scale business with fewer resources.

[3] https://www.microsoft.com/sharepoint
[4] https://www.m-files.com/
[5] https://start.docuware.com/

1.9.2 Enterprise & Regulated Industry Solutions

Hyland (Alfresco / Nuxeo)

Hyland's Alfresco platform [6] is a cloud-native ECM platform that leverages open source with support for open standards and APIs, making it easy to customize [web6]. The architecture is modular and provides REST APIs and SDKs. Meanwhile, the Nuxeo platform provides a highly scalable, low-code, API-first environment. It uses established third-party technologies to undertake activities such as optical character recognition (OCR), speech-to-text, and primary classifications.

Open Text (Documentum / Extended ECM)

Built for extreme demands, Open Text [7] can handle billions of documents with trusted security and flexible cloud deployment options. It features an AI content assistant that provides insights for faster, data-driven decisions.

1.9.3 Open-Source & Technical Solutions

Mayan EDMS

Mayan EDMS [8] focuses on storing, introspecting, and categorizing files while preserving contextual and business information [web7]. It features OCR capabilities, a workflow system, role-based access control, and a robust REST API.

LogicalDOC

LogicalDOC[9] is a web-based DMS that manages popular office formats such as PDF, Word, and Excel [web8]. It supports versioning, messaging, and keyword tags. As a server-side application, it is usable by multiple users simultaneously via common web browsers without requiring local software installation.

[6] https://www.hyland.com/en/resources/alfresco-supported-platforms
[7] https://www.opentext.com/
[8] https://www.mayan-edms.com/
[9] https://www.logicaldoc.com/

1.10 Comparative Analysis

Proprietary solutions such as SharePoint, M-Files, DocuWare, and OpenText provide advanced capabilities including artificial intelligence features, automatic document classification, and very high security standards. However, their major drawbacks are the high licensing and maintenance costs as well as strong vendor lock-in.

Open-source alternatives such as Mayan EDMS, Alfresco, and LogicalDOC offer significant flexibility, substantially lower total cost of ownership (TCO), and a high degree of customizability. These solutions are especially well-suited for academic environments, research projects, and organizations with limited budgets.

Mayan EDMS stands out thanks to its simplicity, built-in OCR (Optical Character Recognition) support, and modern RESTful API, making it particularly appropriate for custom-developed applications and integrations. LogicalDOC provides a user-friendly web interface and strong support for collaborative work. Alfresco is extremely powerful and scalable but requires significantly more effort during installation and configuration.

Solution	**Type**	**Key Advantages**	**Main Drawbacks**
SharePoint	Proprietary	Deep integration with Microsoft 365, high security, advanced AI capabilities	High cost, strong dependence on Microsoft ecosystem
M-Files	Proprietary	Metadata-driven architecture, fast intelligent search, strong automation	Expensive licensing, complex initial configuration
DocuWare	Proprietary	Intelligent automation, high-performance OCR, powerful workflow engine	Very high cost, requires significant user training
Alfresco	Open Source	Highly customizable, REST API, excellent scalability	Complex installation and configuration
Mayan EDMS	Open Source	Free / zero licensing cost, integrated OCR, REST API, easy to deploy	Interface feels somewhat dated
LogicalDOC	Open Source	Simple web interface, good collaboration, multi-format support	Advanced enterprise features are limited

Table 1.1: Comparison of Document Management Solutions

Conclusion

The transition from a traditional paper-based workflow to an electronic Document Management System (DMS) allows organizations to reduce operational costs, automate repetitive processes, and significantly increase document security. This chapter has established the theoretical foundation for our solution by exploring the core components of document capture, storage, and retrieval.

Through a State of the Art review, we analyzed existing market solutions and determined that while proprietary systems like SharePoint, M-Files, and DocuWare offer advanced features, they involve high licensing costs and significant vendor lock-in. Conversely, open-source alternatives such as Mayan EDMS and LogicalDOC provide the necessary flexibility and built-in OCR capabilities required for efficient, searchable data extraction without the burden of expensive licensing.

The objective of this project is to design and implement a cross-platform DMS tailored to modern requirements, leveraging Docker for containerized deployment, OCR to transform scanned images into actionable electronic data, and Elasticsearch to enable rapid, highly scalable full-text searching across the entire repository. By integrating these technologies, the proposed system provides an efficient mechanism to not only manage and share documents, but to instantly retrieve critical information while ensuring data integrity. Building upon this theoretical study, the next chapter will transition to the technical design and conception, detailing the system architecture, use cases, and database schema.

Chapter 2

System Design

Introduction

This chapter translates system requirements into a practical design. We outline the high-level architecture and utilize UML to model the application's structural and dynamic behaviors.

2.1 System Architecture

As shown in Figure 2.5, the EDMS utilizes a multi-tenant SaaS architecture. The SaaS Management Layer handles global platform onboarding, while the Tenant Administration Module manages user provisioning and RBAC. The core engine (*Noyau*) executes document ingestion, *OCR* extraction, and *Elasticsearch* indexing, supported by a decoupled Tenant Blob Storage system for data isolation.

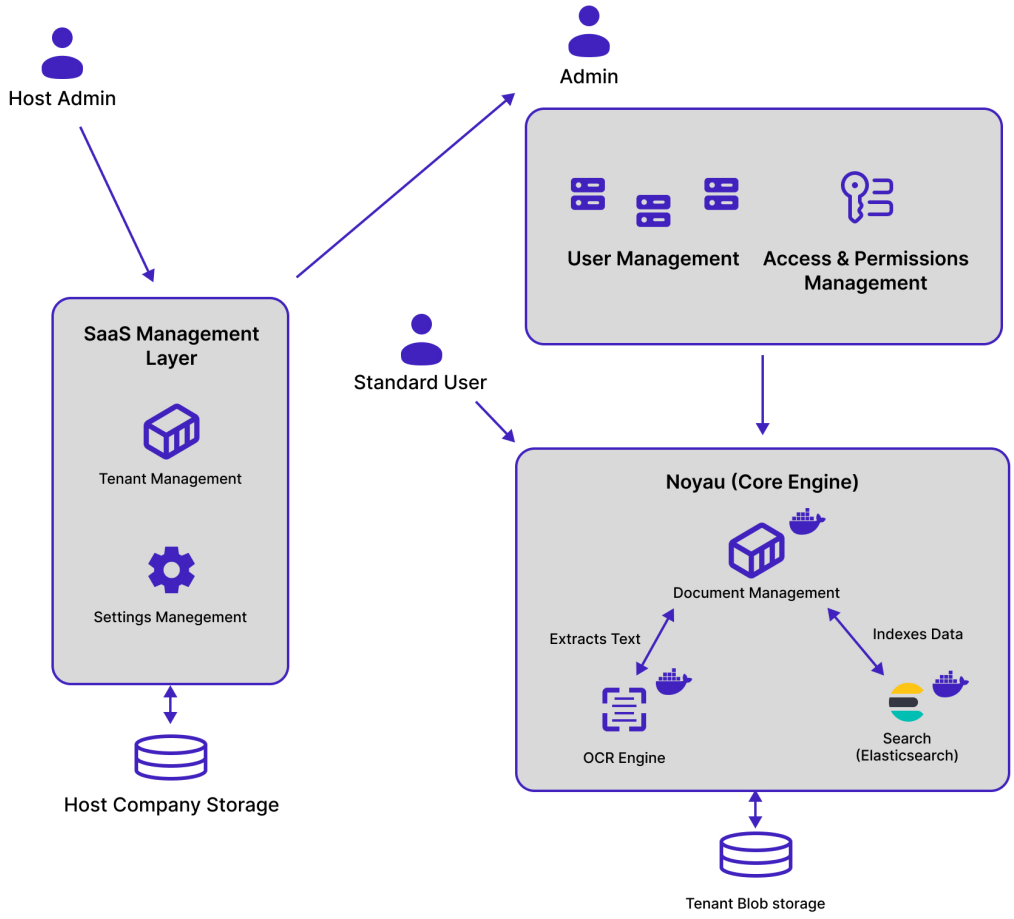


Figure 2.5: Global Architecture Overview of the Document Management System

2.1.1 Description of Architecture Components

To understand the interaction between the different logical building blocks, the role and purpose of each module within the global architecture is detailed below:

- **User Interface (Frontend):** Developed using React and TypeScript, this module acts as the presentation layer. Its purpose is to deliver a smooth and responsive Single-Page Application (SPA), allowing users to browse the workspace, upload files, and configure custom metadata.
- **SaaS Management Layer:** Responsible for managing the platform's software-as-a-service lifecycle, this module handles the dynamic provisioning of client subscriptions (tenants) and configures isolated tenant-specific settings.
- **User Management & Security:** Bound to the 3D security matrix, this component orchestrates user authentication (via JWT tokens), specific tenant subdomain routing, and enforces global Role-Based Access Control (RBAC).
- **System Core Engine:** Developed with the ABP Framework, this core business layer implements Domain-Driven Design (DDD) principles to coordinate document lifecycles, logical data separation via `TenantId`, and secure sharing workflows.
- **OCR (Tesseract) & Search (Elasticsearch) Engines:** The OCR module asynchronously extracts raw text from images and PDF attachments, while Elasticsearch indexes this data to provide instant full-text search and query capabilities across the document repository.
- **Data Engines (PostgreSQL & Object Storage):** PostgreSQL handles relational data persistence and custom metadata schemas utilizing its native `jsonb` support, while Object Storage securely stores the physical file blobs.
- **Containerized Execution Environment (Docker):** Docker serves as the structural foundation by isolating each application block (Core, OCR, Elasticsearch, Database) into independent containers. This guarantees full consistency across environments and simplifies system scalability. This strict isolation allows for rapid deployments and smooth maintenance operations without causing service interruptions for active tenants. Furthermore, this containerized approach prepares the infrastructure for a seamless transition toward production orchestration platforms such as Kubernetes.

2.2 System Modeling

We utilize UML to visualize system behavior and interactions based on our Role-Based Access Control (RBAC) structure.

2.2.1 Use Case Diagram

Figure 2.6 maps the primary use cases for our three main actors:

- **System Admin:** Global configuration and infrastructure management.
- **Manager:** Departmental oversight and permission handling.
- **Standard User:** Core document operations (capture, search, retrieve).

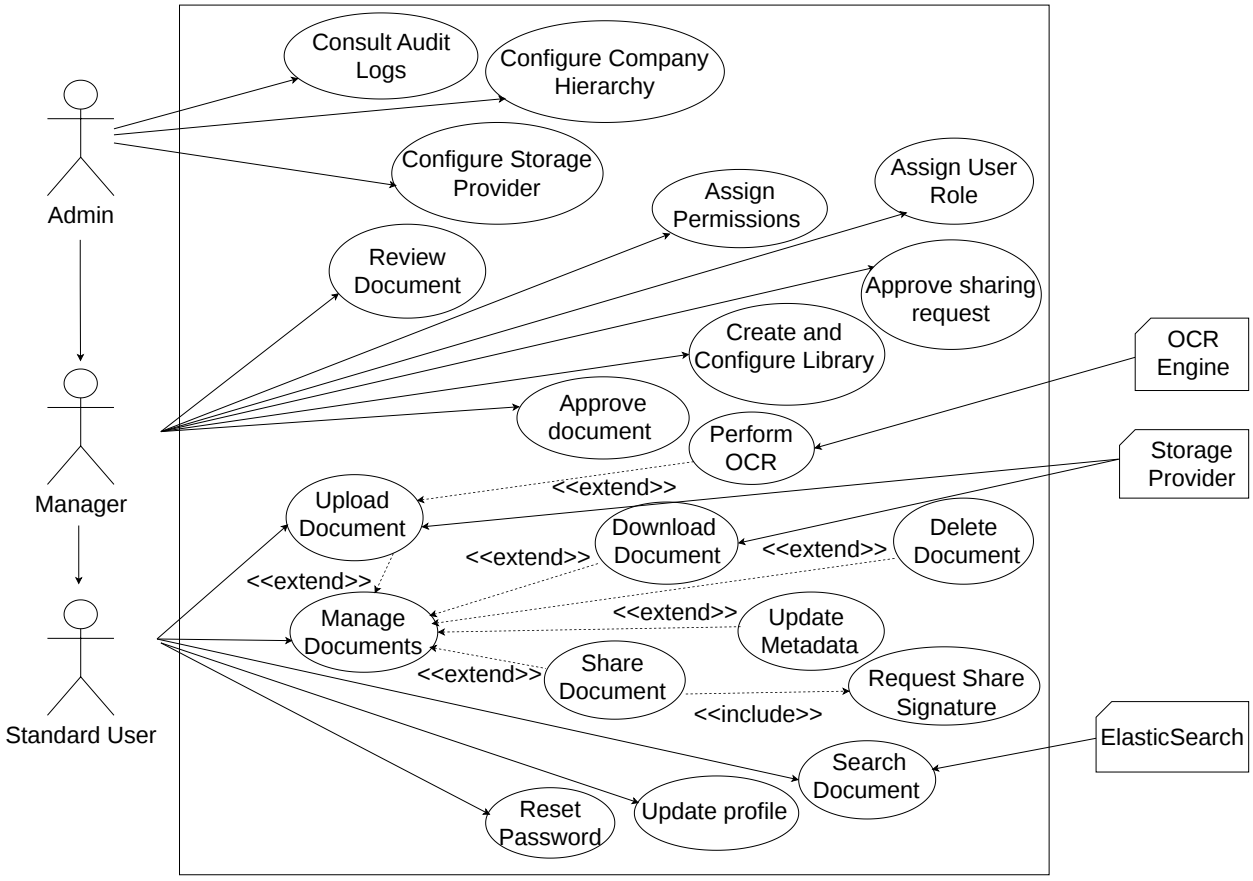


Figure 2.6: Global Use Case Diagram

2.2.2 Activity Diagrams

The following workflows map the operational responsibilities of each user tier:

Standard User Workflow: Covers daily operations such as document uploads, OCR extraction, Elasticsearch querying, and sharing initiation (Figure 2.7).

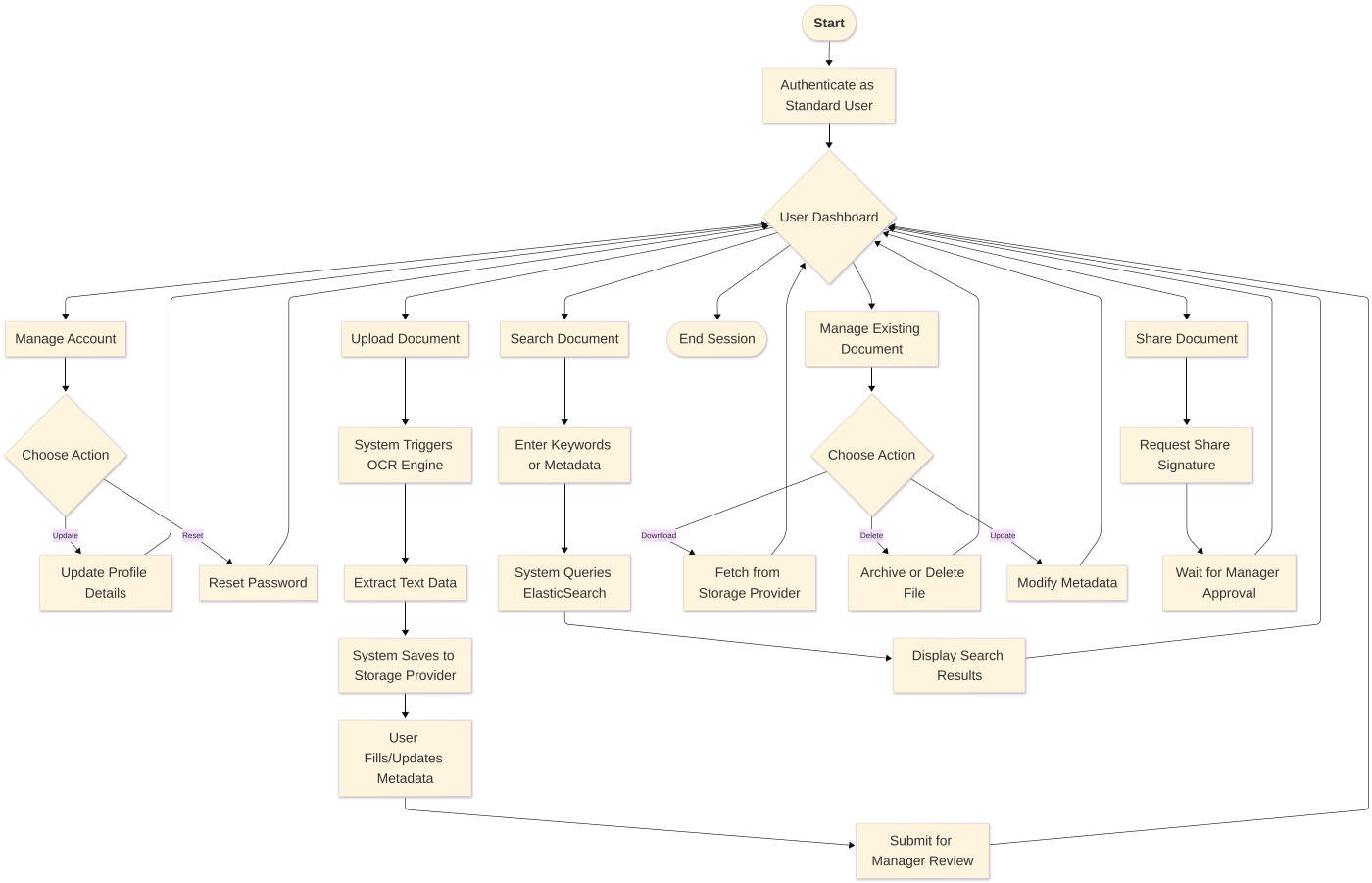


Figure 2.7: User Activity Diagram

Managerial Workflow: Oversees content lifecycles, including workspace provisioning, permissions, sharing requests, and quality control (Figure 2.8).

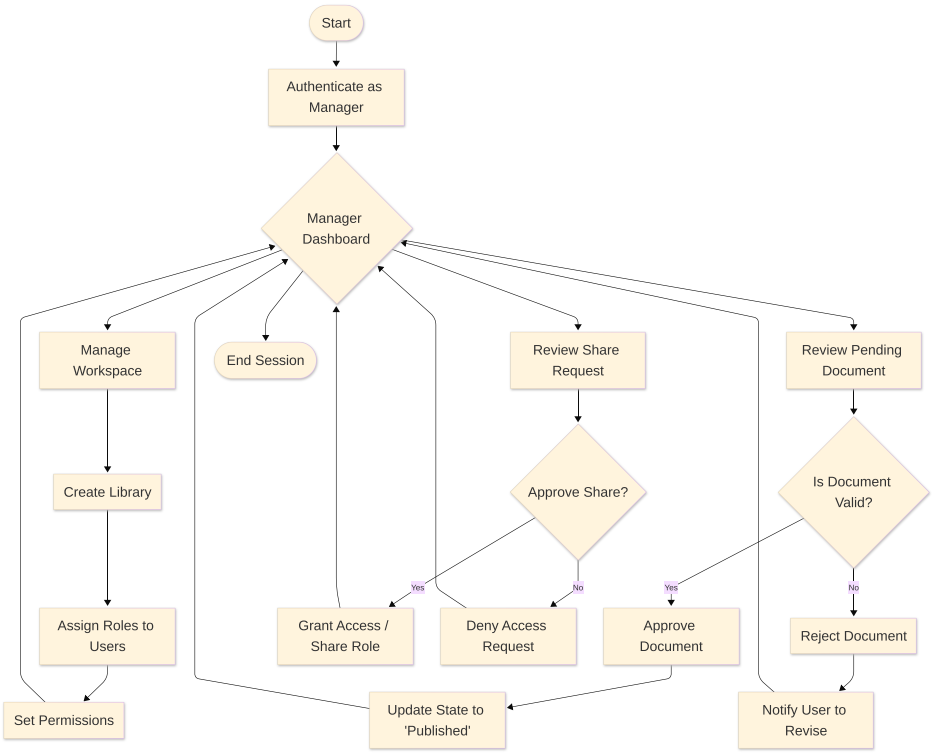


Figure 2.8: Activity Diagram: Managerial Content Lifecycles

Administrative Workflow: Manages system infrastructure, storage providers (AWS S3, Azure, Local), organizational hierarchies, and audit logs (Figure 2.9).

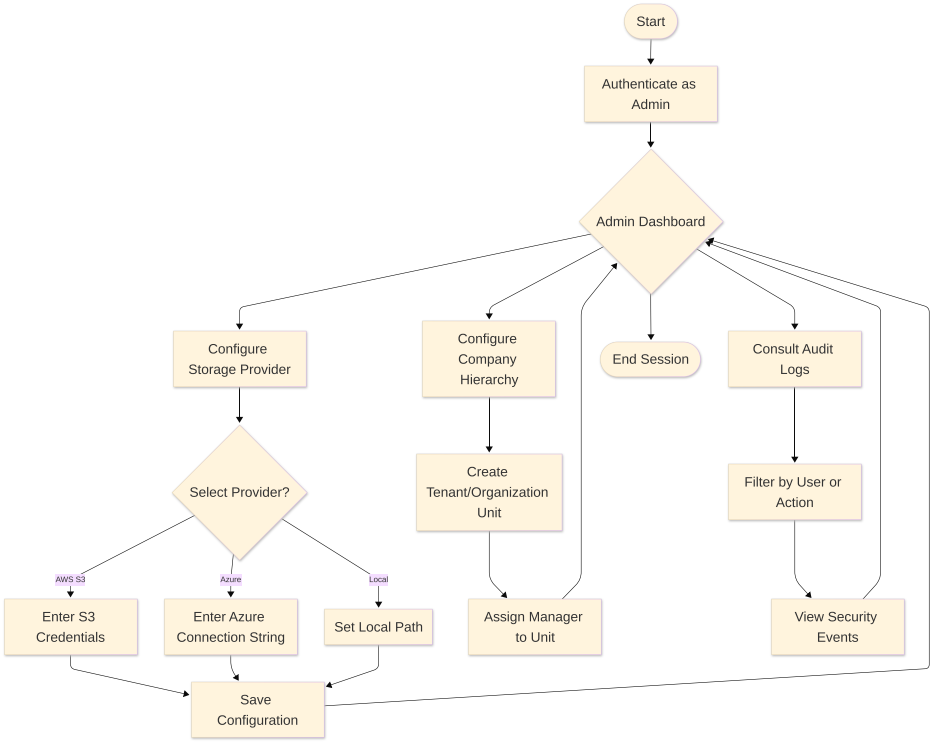


Figure 2.9: Activity Diagram: Administrative Infrastructure

2.2.3 Class Diagram

The class diagram (Figure 2.10) models the static structure of the Electronic Document Management System (EDMS). To manage complexity and ensure high cohesion, the system's domain model is partitioned into four decoupled modules:

- **Documents and Workflow:** This core module governs file lifecycles and organization. `ElectronicDocument` instances are nested within `Folder` entities, which reside inside a parent `Library`. To ensure standardization across the platform, `LibraryTemplate` and `DocumentTemplate` provide reusable architectures. The `Metadata` class implements a dynamic Entity-Attribute-Value (EAV) model to map custom fields. Furthermore, it tracks file lifecycles via the `DocumentState` enumeration and handles collaboration approvals through `documentShareRequest` and `documentShare`.

- **Identity and Access Management:** This module implements multi-tenant SaaS isolation. The `Tenant` class acts as the root boundary, ensuring that all data is strictly scoped to a specific organization. A `User` (linked to a visual `UserProfile`) accesses the system based on clearance levels, with access rights dynamically enforced through the `Role` class to maintain Role-Based Access Control (RBAC).

- **Infrastructure and Storage:** To ensure vendor neutrality and scalability, this module utilizes the **Strategy Design Pattern**. The business logic interacts exclusively with the `StorageProvider` interface. A `StorageManager` orchestrates operations based on `StorageConfig` rules, dynamically routing file bytes to concrete cloud or local implementations like `AzureBlobProvider`, `GoogleCloudProvider`, `S3Provider`, or `LocalDiskProvider`.

- **Traceability and Compliance:** To meet enterprise security standards, the system relies on an **Event-Driven Architecture**. The `IEventPublisher` interface asynchronously broadcasts `SystemEvent` payloads (containing timestamps and context data). An `AuditEventListener` captures these events to generate immutable `AuditLog` records, ensuring a permanent trail of all user actions across the system.

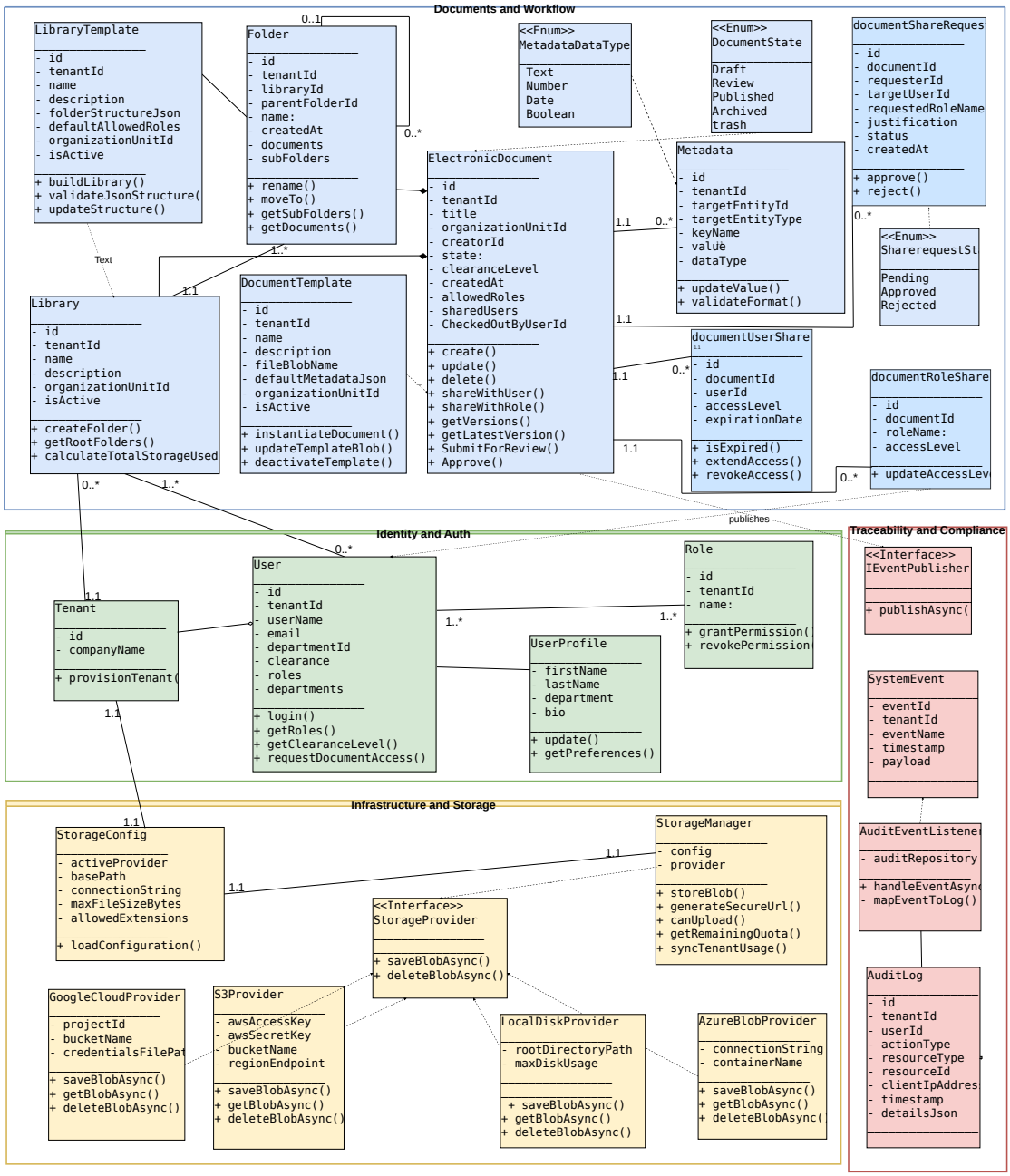


Figure 2.10: Detailed UML Class Diagram of the Document Management System

Conclusion

This chapter established the foundational architecture and dynamic models of the EDMS. By decoupling core modules and defining clear user workflows, we have created a robust blueprint that directly guides the technical implementation phase discussed in the next chapter.

Chapter 3

Implementation

Introduction

In this chapter, we transition from the system design and modeling phase to the practical implementation and realization of our Electr onic Document Management System (EDMS). We begin by establishing the development environment parameters, detailing the hardware configuration, software utilities, and the selected enterprise technology stack. Following this, we look into the core technical implementation features, demonstrating how data isolation, the 3D Security Matrix, and asynchronous text extraction pipelines are built. Finally, we provide a structured visual walkthrough of the system interfaces, illustrating the functional software modules delivered for production environments.

3.1 Development Environment

To ensure code reliability, continuous optimization, and frictionless container deployment, a standardized development environment was provisioned. This section outlines the physical foundation and technical stack configurations used to write and compile the application.

3.1.1 Hardware and Software Tools

The implementation and containerization routines were carried out under the following operational environments:

- **Hardware Infrastructure:**
- **Integrated Development Environments (IDEs):** Visual Studio was chosen to build, debug, and govern the backend web services, whereas Visual Studio Code was used as the lightweight workspace for constructing the modular user interface components.
- **Version Control and Workspace Provisioning:** Git paired with remote repositories managed source history and concurrent branch code streams. Docker and Docker Compose were integrated to maintain isolated local instances of databases, caching caches, and document search clusters.

3.1.2 Technology Stack

To establish complete digital independence for ITComp and guarantee scalable services for its extensive multi-tenant client base, an enterprise-grade architectural stack was selected:

- **Backend Core Framework:** Built on ASP.NET Core combined with the enterprise **ABP Framework**. The ABP Framework handles foundational infrastructural patterns, offering structural modularity, built-in Domain-Driven Design (DDD) layers, and out-of-the-box tenant resolution engines.
- **Frontend User Interface:** A **React** SPA built with **TypeScript** and **Vite** for optimized performance and type safety. It utilizes **React Router** for seamless navigation and **Tailwind CSS** for a responsive, utility-first design. A custom **RBAC layer** dynamically governs UI visibility and feature access based on defined user roles (Admin, Manager, Standard User).
- **Data Engines:** Relational data is managed via PostgreSQL and Entity Framework Core to enforce multi-tenant schemas and metadata. **Elasticsearch** operates as a distributed cluster, providing high-speed search and indexing for text extracted by the OCR engine.
- **Processing Libraries:** The system leverages the **Tesseract OCR** open-source engine to handle text extraction. This engine is integrated within **decoupled execution threads** (Background Jobs), ensuring that the intensive process of converting raw digital images into searchable data matrices does not impact the responsiveness of the main application thread.

3.2 System Interfaces (Presentation of the Solution)

Following a sequential user-journey methodology, this section details the deployment results of the engineered EDMS application via structural software interfaces.

3.2.1 SaaS Host Administration Console

To monitor infrastructure state across the multi-tenant landscape, the system features a global administrative gateway. As shown in Figure 3.11, this high-level interface provides super-administrators with real-time consolidated metrics tracking total tenant accounts, active users, total documents, and cumulative global storage consumption. Additionally, the dashboard isolates specific telemetry details per tenant—such as sub-allocations for *acme*, *djezzy*, *iot*, and *itcomp*—validating strict logical data separation and unified workspace tracking at the host infrastructure layer.

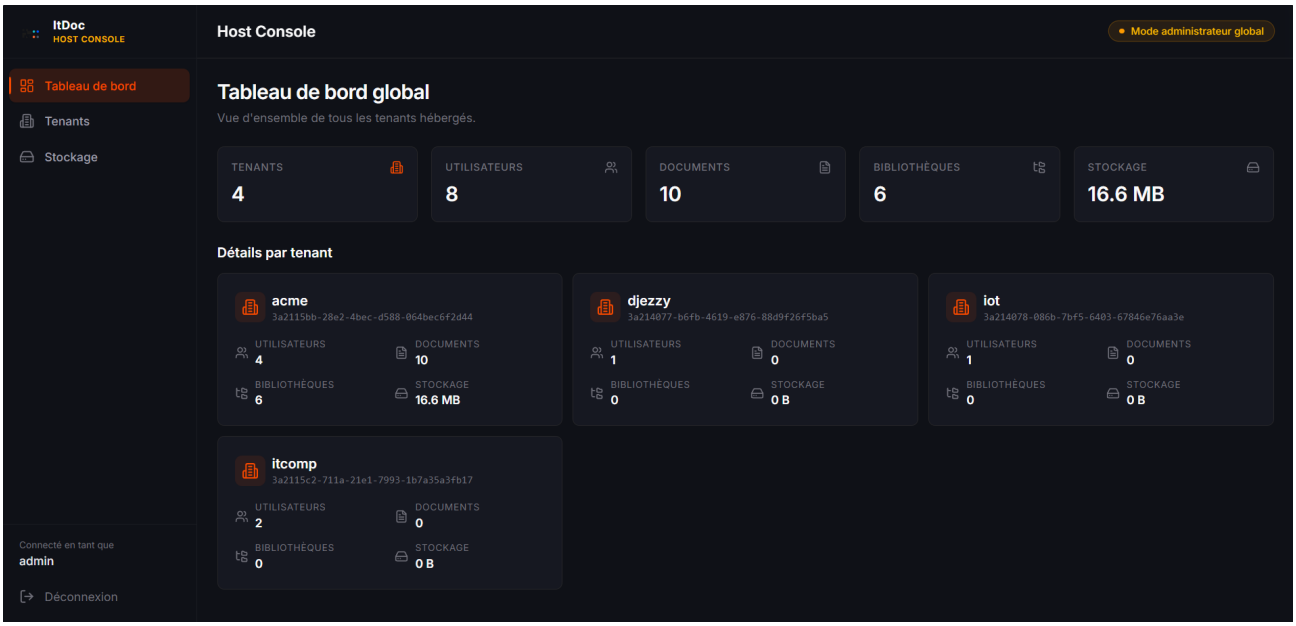


Figure 3.11: Host Console Interface displaying Global Telemetry and Per-Tenant Metrics

3.2.2 Authentication and Multi-Tenant Portal

The entry module serves as a secure gateway, identifying user spaces before evaluating login profiles. As shown in Figure 3.12, the portal resolves tenant scopes through web subdomains, validating security identities and issuing signed JSON Web Tokens (JWT) upon authorization.

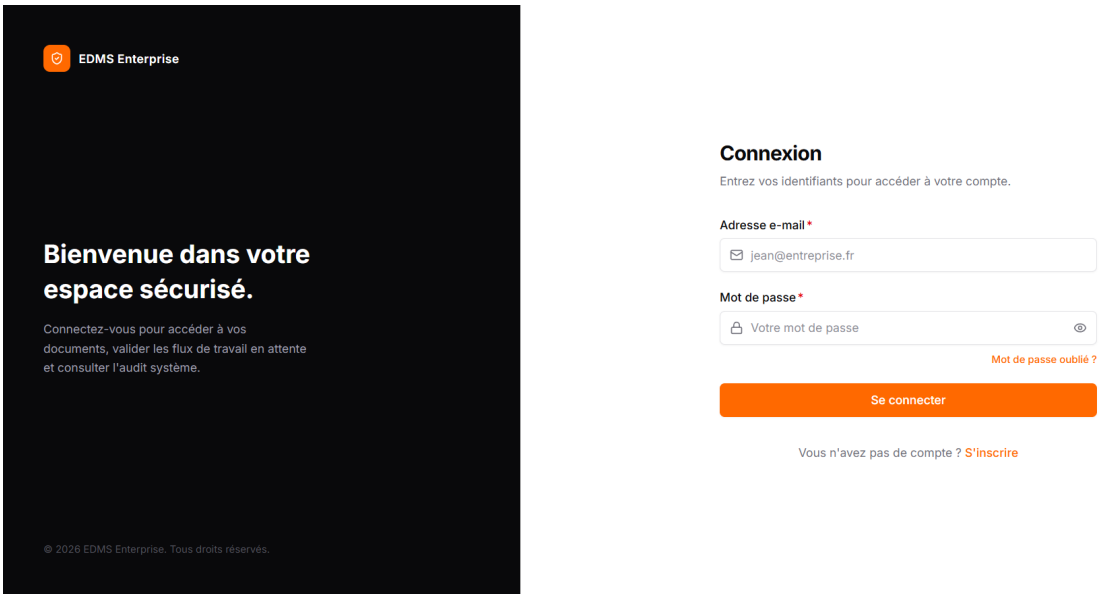


Figure 3.12: System Authentication Interface with Dynamic Tenant Subdomain Verification

3.2.3 Tenant Dashboard

Following authentication, users are redirected to customized operational hubs. Figure 3.13 illustrates the primary landing dashboard view, displaying current storage utilization metrics, system activity logs, and active operational departments.

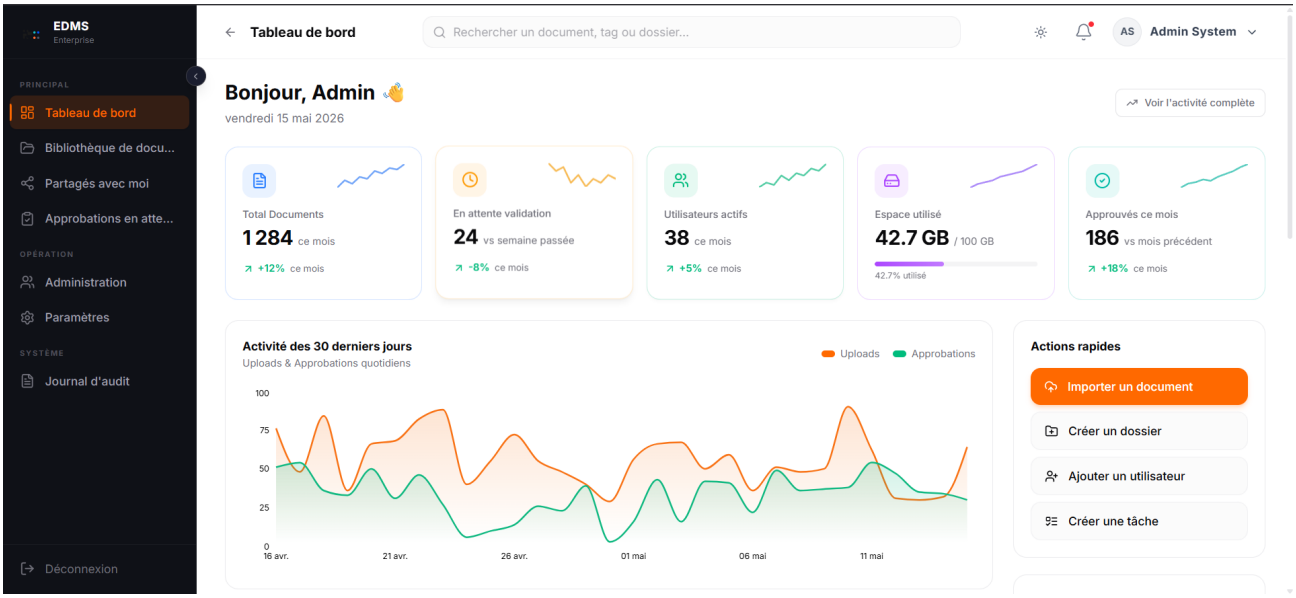


Figure 3.13: Workspace Dashboard with Multi-Tenant Storage Analytics

3.2.4 Library Template Provisioning and Workspace View

As illustrated in Figure 3.14, the workspace rendering engine displays dynamically provisioned organizational library spaces.

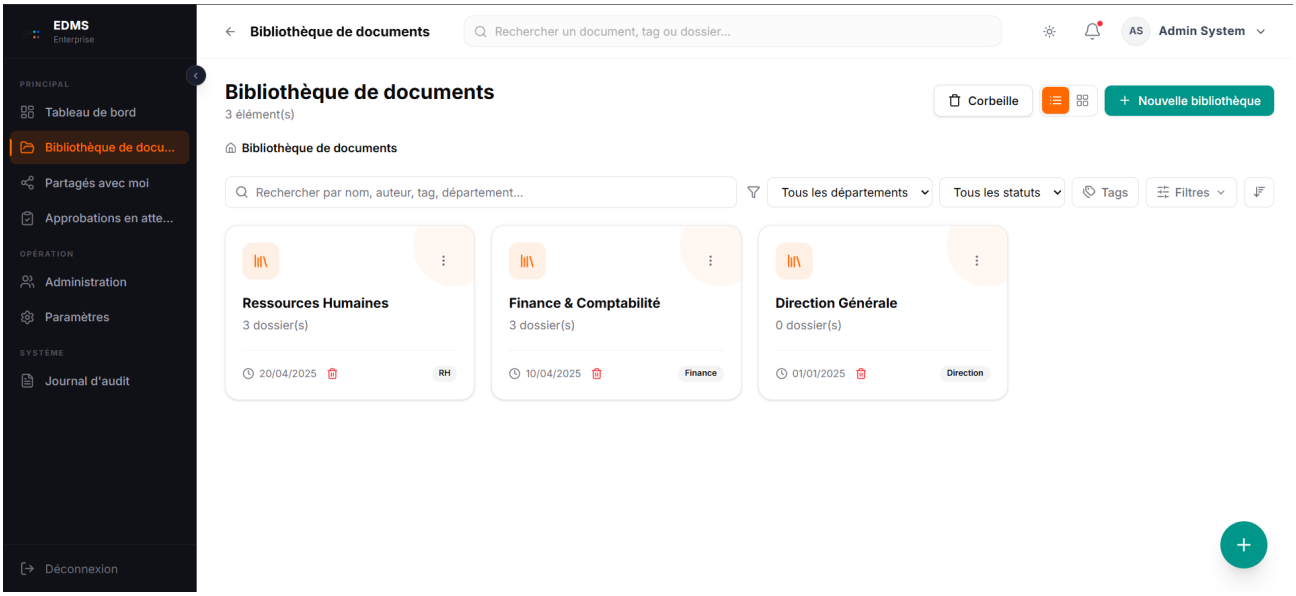


Figure 3.14: Dynamic Library Template View with Multi-Tenant Workspace Segmentation

3.2.5 Advanced Document Explorer

The document explorer handles workspace navigation and file management. As shown in Figure 3.15, this interface provides a custom tags, alongside an asynchronous menu for file uploads, scanning, and directory provisioning.

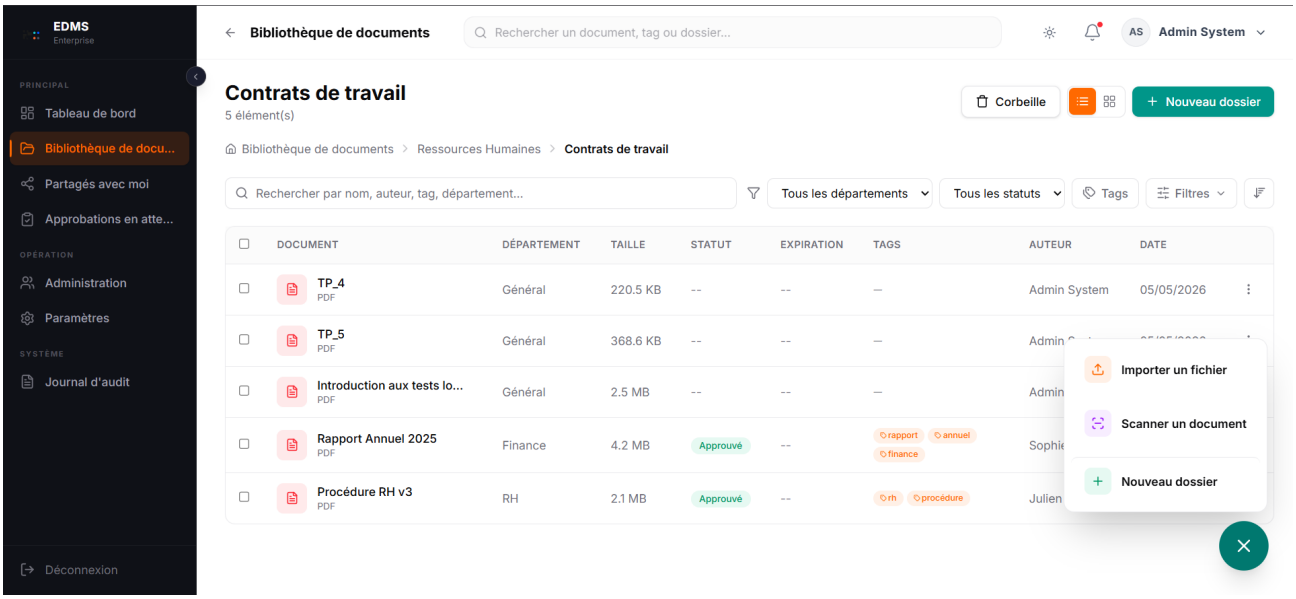


Figure 3.15: Document Explorer Interface with Metadata Grid and Action Menu

3.2.6 Elasticsearch Full-Text Search

The search system relies on an Elasticsearch cluster to provide high-speed information retrieval. As shown in Figure 3.16, this interface allows users to execute complex queries across the entire tenant workspace, indexing and searching both structured metadata properties and unstructured content extracted by the Tesseract OCR engine.



Figure 3.16: Elasticsearch Search Interface for Content and Metadata Retrieval

3.2.7 Dynamic Metadata Template Builder

To allow advanced classification beyond static folders, the system provides a dynamic schema engine. As shown in Figure 3.17, this interface enables administrators to design custom metadata templates (e.g., expiration dates, invoice numbers) for specific departments, enforcing uniform data profiling across the tenant's repository.

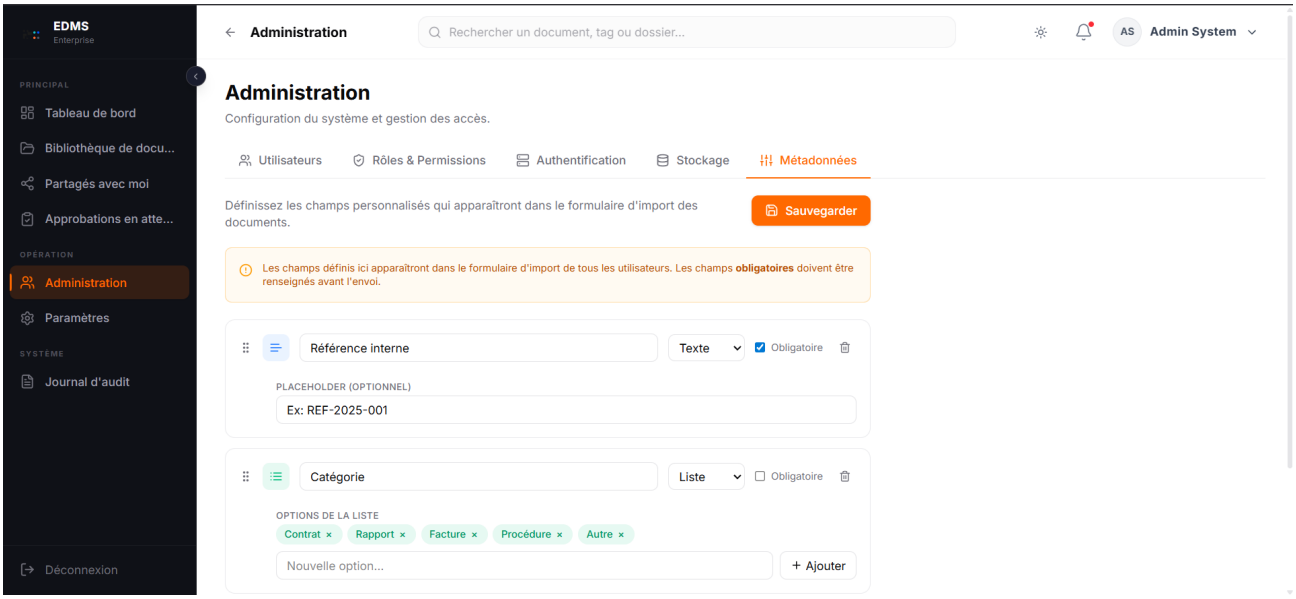


Figure 3.17: Dynamic Metadata Template Builder Interface

3.2.8 Asynchronous Intelligent Ingestion and OCR Scanner

The document ingestion pipeline uses an AI-powered digitization interface to process physical files. As illustrated in Figure 3.18, when an unindexed document (such as an administrative form) is uploaded, the integrated Tesseract OCR engine initiates an automated text extraction sequence. This multi-phase pipeline extracts unstructured content directly from the scanned image matrix, preparing the raw text data for metadata field mapping and full-text index provisioning within the search cluster.

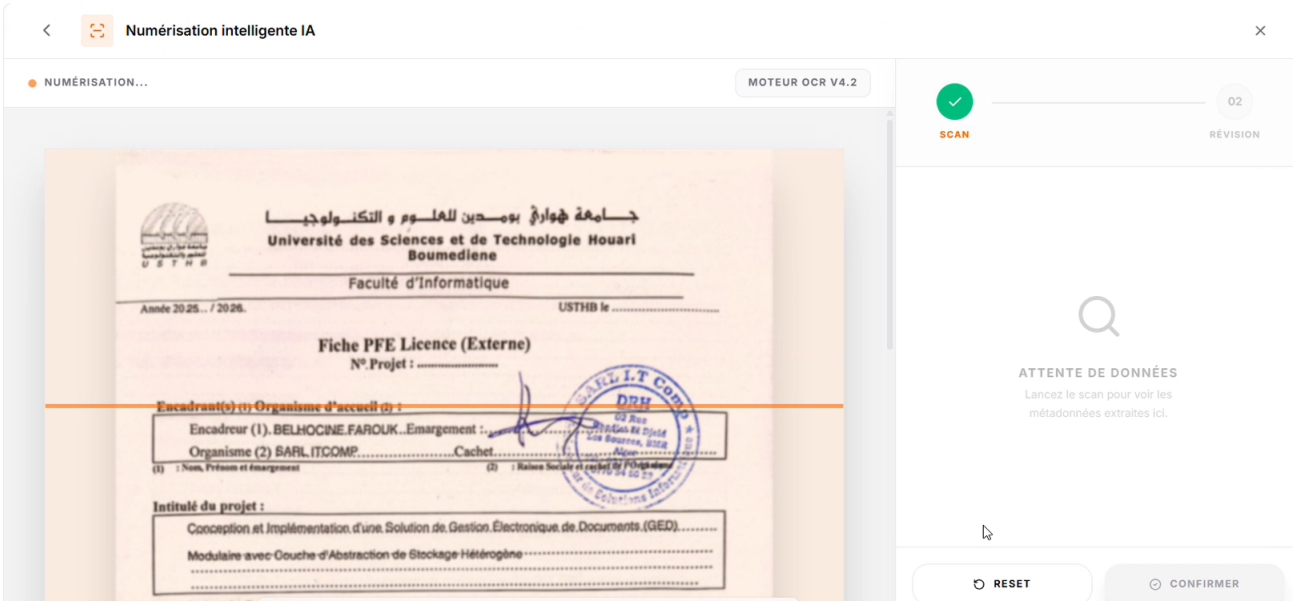


Figure 3.18: Intelligent Scan Interface with Integrated OCR Document Processing

Conclusion

In this chapter, we documented the implementation and practical realization of the Electronic Document Management System (EDMS). By utilizing modern framework principles through .NET Core, React, and the ABP Framework, we achieved multi-tenant logical partitioning, background asynchronous text extraction queues, and advanced data tracking architectures. This deployment eliminates licensing dependency bottlenecks for ITComp and ensures total alignment with localized personal data safety guidelines (such as Algerian Law 18-07). With the system fully functional, it is now positioned for systematic validation and testing evaluations.

General Conclusion

This project has materialized the design and implementation of a multi-tenant Electronic Document Management System (EDMS) delivered as a secure Software as a Service (SaaS). In response to the growing complexity of enterprise data sprawl and the critical need for strict information governance, our solution brings significant value by digitizing physical workflows, automating document categorization, and ensuring secure, highly scalable storage for diverse organizational units.

The implementation relied on a robust Domain-Driven Design (DDD) architecture built upon the ABP Framework and PostgreSQL, seamlessly integrating containerized microservices managed via Docker. This approach allowed for optimal scalability and absolute multi-tenant data isolation. Furthermore, the integration of a flexible Strategy Pattern for storage orchestration ensured seamless compatibility with major cloud providers (AWS S3, Azure Blob) as well as local infrastructure, guaranteeing both high availability and adaptability to varying client requirements.

Thanks to the integration of an Optical Character Recognition (OCR) engine and Elasticsearch, the tool offers a comprehensive and automated capability for document capture, intelligent metadata extraction, and rapid full-text retrieval. The user workflow—from secure authentication governed by a strict 3D security matrix (RBAC and ACLs), to dynamic workspace provisioning via JSON templates, and automated form-to-document generation—has been designed to be highly intuitive and efficient. The ability to seamlessly manage access requests and coordinate team sharing underscores the practical, collaborative, and operational aspect of the solution.

In summary, the developed system represents a tangible advancement in enterprise content management, offering organizations a highly secure, high-performance, and maintainable platform for organizing, protecting, and collaborating on their most critical digital assets.

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